GRADE 10 REVISION: TRIGONOMETRY

5.4 $T = \frac{3.\sin(x+20^\circ)}{\cos y - \tan 40^\circ}$ Use a calculator to calculate the value of T to 3-decimal places if:

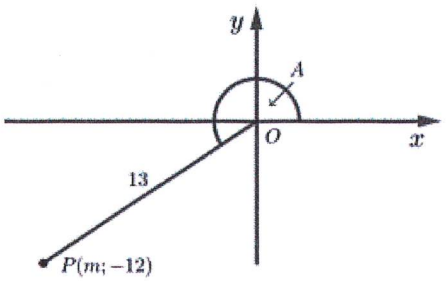
5.4.1 $x = 35,8^\circ$ and $y = 60^\circ$ (2) 5.4.2 $x = 130^\circ$ and $y = x + 20^\circ$ (3)

6.2 Given that $25.\sin A + 7 = 0$ and $\cos A > 0$, without using a calculator, determine the value of:

6.2.1 $\cos A$ (3) 6.2.2 $\sin^2 A + (\cos A)^2$ (3)

6.2.4 $1 + \tan A - \frac{1}{\sin A}$ (3)

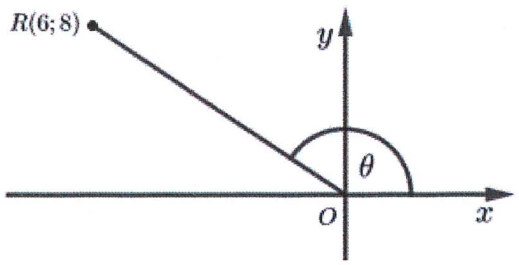
6.3 In the diagram, $OP = 13$ with $P(m\,;-12)$ and $x\hat{O}P = A$.



Without using a calculator, determine the value of:

6.3.1 m (2) 6.3.2 $\cos A$ (2)

6.4 In the diagram $R(6\,;8)$ lies on the line OR. $x\hat{O}R = \theta$.



Without using a calculator, determine the value of:

6.4.1 the length OR. (2) 6.4.2 $1 - 2\sin\theta \times \cos\theta$ (3)

6.4.3 $\cos 2\theta$ if $\cos 2\theta = \cos^2\theta - \sin^2\theta$ (3)

6.6 If $-tanA = \frac{4}{3}$ and $0^\circ \leq A \leq 180^\circ$, without the use of a calculator, determine the value of the following:

6.6.1 $sinA$ (3)

6.7 Given that $B \in [0^\circ ; 360^\circ]$, which quadrant(s) will/can angle B lie in if:

6.7.1 $sinB > 0$ and $cosB < 0$? (2)

6.7.2 $cosB > 0$ and $tanB < 0$? (2)

6.7.3 $sinB \times tanB > 0$? (3)

7.1 Solve the following equations for $x \in [0^\circ ; 90^\circ]$:

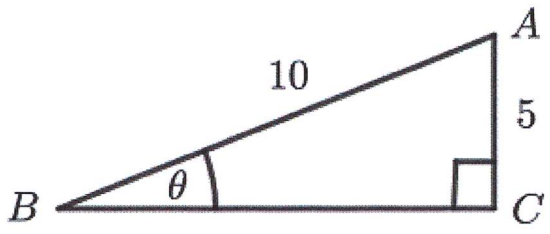
7.1.1 $-3.sin2x + 1 = 0$ (2)

7.1.2 $tan\left(\frac{1}{2}x + 40^\circ\right) = 4$ (2)

7.1.3 $\sqrt{5}\,cos(x - 5^\circ) = 2$ (2)

7.1.4 $4.sin\frac{x}{3} = \sqrt{3}$ (3)

8.1 Right-angled triangle ABC is shown in the figure.



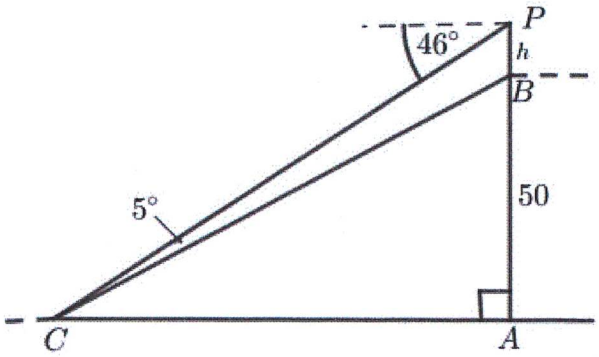
<u>Without the use of a calculator and showing all working</u>:

8.1.1 determine the value of θ. (2)

8.1.2 hence use θ to calculate the length BC <u>in simplest surd form</u>. (3)

8.1.3 determine BC by applying Pythagoras's Theorem <u>in simplest surd form</u>. (2)

8.2 In the sketch, CA is the horizontal ground level. The side AB of a building is perpendicular to CA with $AB = 50$ metres. PB is a vertical pole height h and $B\hat{C}P = 5^\circ$. The angle of depression of C from P is 46°.



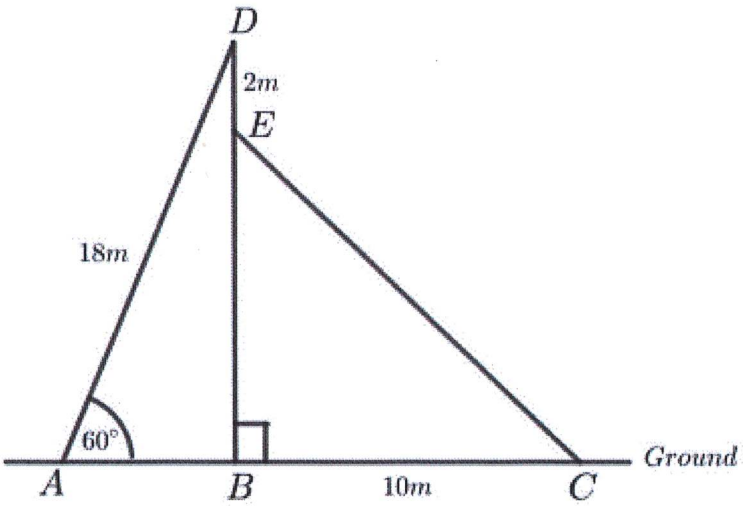
Determine the value of h. (6)

8.4 In the diagram, A, B, C, D, and E all lie in the same vertical plane. ABC is the horizontal ground level. BD is a vertical flagpole which is supported by two steel cables, AD and CE. The angle of elevation of D from A is 60°.
$AD = 18\ metres$; $BC = 10\ metres$; $DE = 2\ metres$

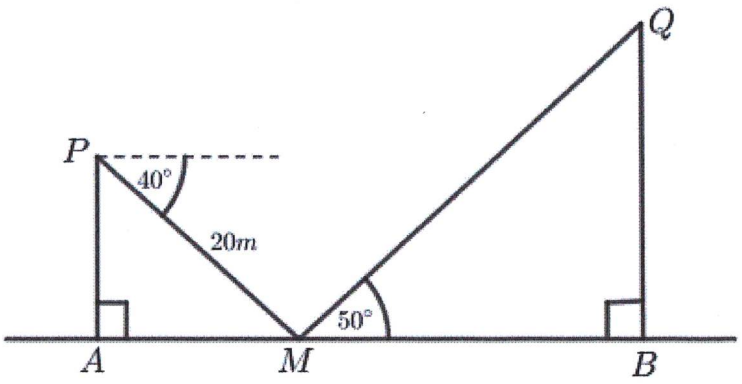
8.4.1 Calculate the height of the flagpole BD, leaving your answer in simplest surd form) (2)

8.4.2 Hence calculate the length of the steel cable CE. (4)

8.4.3 Determine the angle of elevation of E from C, correct to ONE decimal place. (2)



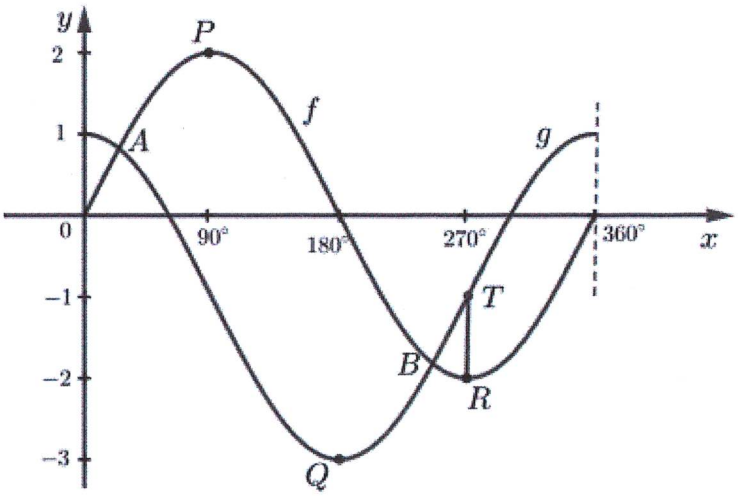
8.7 In the figure AMB is the horizontal ground level. AP and BQ are vertical poles with $AP = \frac{1}{2}BQ$. $PM = 20$ metres. A, M, B, P and Q all lie in the same vertical plane. The angle of depression of M from P is 40° and the angle of elevation of Q from M is 50°.



8.7.1 Determine the distance AB. (7)

8.7.2 Determine the distance PQ. (6)

9.4 In the diagram the graphs of $f(x) = m.sinx$ and $g(x) = 2.cosx + n$ are shown for $x \in [0^\circ; 360^\circ]$. $Q(180^\circ; -3)$ is a turning point on g. P and R are turning points on f. The two graphs intersect at $A(24{,}30^\circ; 0{,}82)$ and $B(245{,}70^\circ; -1{,}82)$.



9.4.1 Write down the values of m and n. (2)

9.4.2 Write down the coordinates of P and Q. (2)

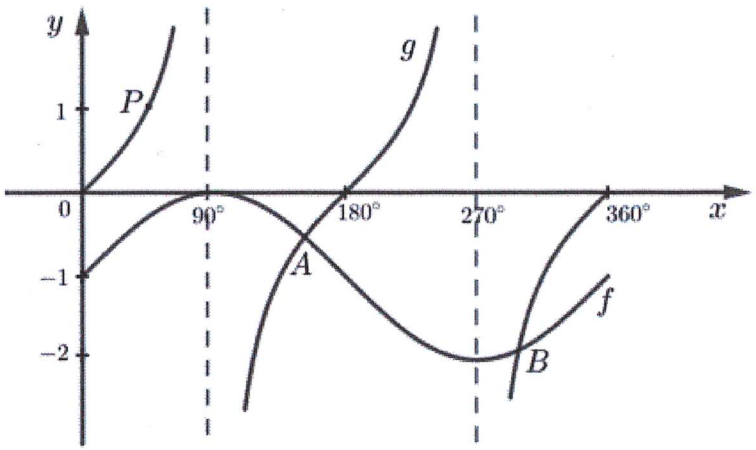
9.4.3 $TR \parallel y$-axis with T on g. Determine the length TR. (2)

9.4.4 Determine the coordinates where $g(x) = 0$. (3)

9.4.5 For which values of x is:

9.4.5.1 $f(x) \leq 0$? (2) 9.4.5.2 $g(x) > f(x)$? (2) 9.4.5.3 $f(x).g(x) > 0$? (4)

9.5 In the sketch, $f(x) = sinx + m$ and $g(x) = tanx$ are shown for $x \in [0° ; 360°]$. $P(n ; 1)$ lies on g. A and B are the points of intersection of the two graphs with x_A=152,04° and x_B = 297,97°.



9.5.1 Write down the value of m. (1)

9.5.2 What is the range of $f(x)$? (1)

9.5.3 Write down the value of n. (1)

9.5.4 For which values of x is:

9.5.4.1 $f(x) > g(x)$? (4) 9.5.4.2 $f(x).g(x) = 0$? (3)

9.5.5 $h(x) = f(x) + k$. For which value(s) of k will h have exactly one x-intercept? (2)